

HW Class 6

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```
library(bio3d)
```

Q6: How would you generalize the original code above to work with any set of input protein structures?

This function takes a PDB ID, reads the protein, and plots the B-factor values from chain A alpha carbons. The plot shows which parts of the protein are more flexible and where drug binding could happen. The output of the function is a plot that shows B-factor values across the protein sequence.

```
plot_protein_bfactor <- function(pdb_id) {  
  
  pdb <- read.pdb(pdb_id)  
  chainA <- trim.pdb(pdb, chain = "A", elety = "CA")  
  plotb3(chainA$atom$b, sse = chainA, typ = "l", ylab = "B-factor")  
}
```

```
plot_protein_bfactor("4AKE")
```

Note: Accessing on-line PDB file

